Foundations of Reinforcement Learning

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Lecture 1: MAB and MDP Basics

1.1 Learning from interaction

A reinforcement learning agent selects actions, receives rewards, and interacts with an environment. The objective is to choose a policy that achieves high cumulative reward. Two central concerns are sample efficiency, the amount of experience required, and computational efficiency. Exploration gathers information; exploitation uses the best action currently known.

- Multi-armed bandit: choose an arm $a \in \mathcal{A}$ and observe a random reward with mean $r(a)$. The objective in the lecture's formulation is to identify a best arm. One example models arm a's rewards as $\mathcal{N}(r(a), \sigma^2)$.
- Contextual bandit: observe a context $s \in \mathcal{S}$, choose an action, and observe its reward. A policy maps each context to an action or an action distribution. The goal is to identify a good action for each context.
- Markov decision process (MDP): actions also affect future states. A good action must account for later rewards, not only its immediate reward.

For exploration, ϵ-greedy chooses a random action with probability ϵ and a currently greedy action otherwise. Upper confidence methods instead compare optimistic estimates of action values, incorporating uncertainty.

1.2 Finite-horizon MDPs

An episodic MDP is specified by $(\mathcal{S}, \mathcal{A}, P, r, H)$: state and action sets, transitions $P_h(s' \mid s, a)$, rewards $r_h(s, a)$, and horizon H. At each step $h = 1, \ldots, H$, the agent observes s_h, samples $a_h \sim \pi_h(\cdot \mid s_h)$, receives $r_h(s_h, a_h)$, and moves to $s_{h+1} \sim P_h(\cdot \mid s_h, a_h)$. The episode begins at s_1 and ends after step H.

A policy π_h can be deterministic or assign a distribution over actions. Its expected cumulative rewards from a state and from a state-action pair are

$$V_h^\pi(s) = \mathbb{E}_\pi\left[\sum_{t=h}^{H} r_t(s_t, a_t) \,\middle|\, s_h = s\right], \quad Q_h^\pi(s, a) = \mathbb{E}_\pi\left[\sum_{t=h}^{H} r_t(s_t, a_t) \,\middle|\, s_h = s, a_h = a\right].$$

An optimal policy maximizes $V_1^\pi(s_1)$ for a fixed start, or its expectation over the initial-state distribution. Expectations include transition and policy randomness.

Example: A robot can take a small immediate reward or move to a state offering a larger later reward. A contextual bandit only evaluates the current context and action; an MDP also models how that choice changes the next state and future opportunities.

1.3 Access to the environment

- **Known model:** rewards and transitions are available, so the task is planning.
- **Simulator:** a generative model can be queried at a chosen state-action pair to sample outcomes.
- **Online interaction:** the model is unknown and experience follows trajectories from the initial state; arbitrary states cannot simply be queried.

1.4 Infinite-horizon discounting

For $0 < \gamma < 1$ and rewards $r_t \in [0, 1]$, the discounted return is $\sum_{t=0}^{\infty} \gamma^t r_t$. Rewards after the first H steps have total at most

$$\sum_{t=H}^{\infty} \gamma^t = \frac{\gamma^H}{1-\gamma}.$$

To bound this tail by $\varepsilon > 0$, it suffices that $H \geq \log((1-\gamma)\varepsilon)/\log\gamma$. Thus the effective horizon grows roughly as $(1-\gamma)^{-1}$ up to logarithmic factors: as γ approaches one, more distant rewards matter.

Source. Chi Jin, Foundations of Reinforcement Learning, Spring 2020, Lecture 1 slides and notes.